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REVIEW ARTICLE

Paracetamol (acetaminophen) poisoning: The early years

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Paracetamol (acetaminophen) was marketed in the 1950s as a nonprescription analgesic/antipyretic without any preclinical toxicity studies. It became used increasingly for self-poisoning, particularly in the UK and was belatedly found to cause acute liver damage, which could be fatal. Management of poisoned patients was difficult as maximum abnormalities of liver function were delayed for 3 days or more after an overdose. There was no treatment and the mechanism of hepatotoxicity was not known. The paracetamol half-life was prolonged with liver damage occurring when it exceeded 4 h and the Rumack–Matthew nomogram was an important advance that allowed stratification of patients into separate zones of risk. It is used to guide prognosis and treatment and its predictive value could be increased by combining it with the paracetamol half-life. The problems of a sheep farmer in Australia in the early 1970s led to the discovery of the mechanism of paracetamol hepatotoxicity, and the first effective treatment of overdosage with intravenous (IV) cysteamine. This had unpleasant side effects and administration was difficult. *N*-acetylcysteine soon became the treatment of choice for paracetamol overdose and given early it was very effective when administered either IV or orally. *N*-acetylcysteine could cause *anaphylactoid* reactions, particularly early during IV administration when the concentrations were highest. Simpler and shorter regimes with slower initial rates of infusion have now been introduced with a reduced incidence of these adverse effects. In addition, there has been a move to use larger doses of *N*-acetylcysteine given over longer periods for patients who are more severely poisoned and those with risk factors. There has been much interest recently in the search for novel biomarkers such as microRNAs, procalcitonin and cyclophilin that promise to have greater specificity and sensitivity than transaminases. Paracetamol–protein adducts predict hepatotoxicity and are specific biomarkers of toxic paracetamol metabolite exposure. Another approach would be measurement of the plasma levels of cysteine and inorganic sulfate. It is 50 years since the first effective treatment for paracetamol poisoning and, apart from liver transplantation, there is still no effective treatment for patients who present late.

KEYWORDS
acetaminophen, glutathione, hepatotoxicity, *N*-acetylcysteine, nomogram, overdose, paracetamol

1 | INTRODUCTION

Paracetamol (acetaminophen) had only brief clinical use as an antipyretic towards the end of the 19th century, but interest was rekindled when it was rediscovered as the major metabolite of the established analgesics acetanilide and phenacetin.[1,2] In 1950, it was marketed in the USA in combination with aspirin and caffeine as Triogesic but was withdrawn a year later following reports of agranulocytosis. No causal relationship was found and it was reintroduced on prescription. It became available over the counter in 1955. It was introduced without prescription in the UK in 1956.[3,4] At that time, aspirin was the most popular antipyretic analgesic but there was increasing concern about its adverse gastrointestinal and haematological effects. It was also suspected as a cause of Reye's syndrome in children.[5] Paracetamol did not produce these side effects and in normal therapeutic doses it was remarkably free of adverse effects. Over the years it replaced aspirin as the most popular nonprescription pain killer/antipyretic in many countries. In the UK, aspirin had long been the most commonly employed agent for self-poisoning, particularly by young adults but now paracetamol was increasingly being taken instead.[6] In the late 1960s and the 1970s the number of admissions to hospital with paracetamol poisoning rose dramatically and cases were reported increasingly from many other countries.[7] No preclinical toxicity studies with paracetamol had ever been carried out and in 1966 it came as a complete surprise to find that it could cause acute severe and fatal centrilobular liver necrosis when taken in overdose.[8] Animal studies confirmed the acute hepatotoxicity of paracetamol.[9,10] In 1970, the clinical description of paracetamol poisoning was reported in a series of 41 Edinburgh patients by Proudfoot and Wright[11] and the first case in the USA was reported in 1971.[12] Acute centrilobular hepatic necrosis was the primary manifestation of paracetamol toxicity following an overdose and renal failure could also occur, usually, but not always, in patients with severe liver damage.

2 | THE EARLY DAYS

At that time there was no effective treatment and clinicians faced particular difficulties in the management of these patients. During the first few hours, there were no specific symptoms or signs and there was no change in the level of consciousness unless a massive dose of paracetamol had been taken or central nervous system depressants had also been ingested. Liver function tests could be normal or only slightly deranged on admission to hospital but several days later the patient could die with fulminant hepatic failure.[8] Unless presentation was considerably delayed, there was no way to tell whether a patient would live or die. To make matters worse, there was great individual variation in susceptibility to paracetamol hepatotoxicity and the mechanism whereby it caused liver damage was unknown. Before effective treatment became available in 1973, <10% of unselected patients admitted to the Edinburgh Royal Infirmary after taking an overdose of paracetamol developed severe liver damage (arbitrarily defined as a plasma aminotransferase activity of 1000 IU/L or more). About 1% progressed to acute hepatic failure which in those pre-liver transplant times had a mortality approaching 100%.[13]

I joined the staff of the Regional Poisoning Treatment Centre (Ward 3) at the old Edinburgh Royal Infirmary in 1969. At that time, the Centre was being run single-handedly by Henry Matthew, who has been referred to as "The father of modern clinical toxicology".[14] When Henry retired in the early 1970s, he was succeeded by Alex Proudfoot. Meantime, there was an urgent need to find some early way to assess the severity of paracetamol poisoning and thus the likely prognosis. Unfortunately, no reliance could be placed on the amount claimed to have been taken and maximum biochemical abnormalities of liver function tests were delayed until 3 or more days after the overdose. Serial plasma concentrations of paracetamol were determined in 30 patients following an overdose and a definitive relationship between paracetamol concentrations, time after ingestion and hepatotoxicity was established (Figure 1). The plasma half-life of paracetamol in adults following a therapeutic dose is about 2 h but after an overdose it increased progressively with the severity of liver damage. A half-life of 4 h or more was associated with liver damage and in patients who progressed to acute liver failure, the conjugation of paracetamol was grossly impaired and the half-life was greatly prolonged.

Plasma paracetamol concentrations could only be interpreted when the time from the overdose was taken into account. Concentrations >300 mg/L at 4 h after the overdose were invariably associated with severe and sometimes fatal hepatotoxicity. Conversely, the risk was very low in patients with 4-h concentrations <100 mg/L. Following an overdose there was a rapid decrease in the sulfate conjugation of paracetamol and this caused a modest increase in the half-life up to 3½ to 4 h. With increasingly severe liver damage there was further prolongation of the half-life beyond 4 h. It is important to note that the paracetamol half-life was prolonged from the time of the earliest observations and it remained relatively constant in the individual patient over the following 24–36 h unless hepatic failure developed in which case the half-life increased progressively up to as long as 60 h.[15,16]

In effect, the paracetamol half-life is a very early *built-in* liver function test and predictor of mortality. A later study confirmed the correlation of the half-life with liver damage in patients treated with *N*-acetylcysteine but a half-life of 5.5 h was said to give a better discrimination between patients with little or no liver damage and those with severe injury. However, in many patients, only 2 plasma paracetamol measurements were made.[17] At least 3 timed measurements of the paracetamol concentration are required over a period of, say, 8 h. An accurate prognosis could then be established well within 12 h of presentation. Importantly, the half-life is independent of the patient's recall of the time of ingestion. There will be problems with staggered overdoses, overdoses of extended-release formulations and occasions when the absorption of paracetamol is delayed for other reasons. In such circumstances the errors would be on the side of safety and there would be no risk of missing severe liver damage.

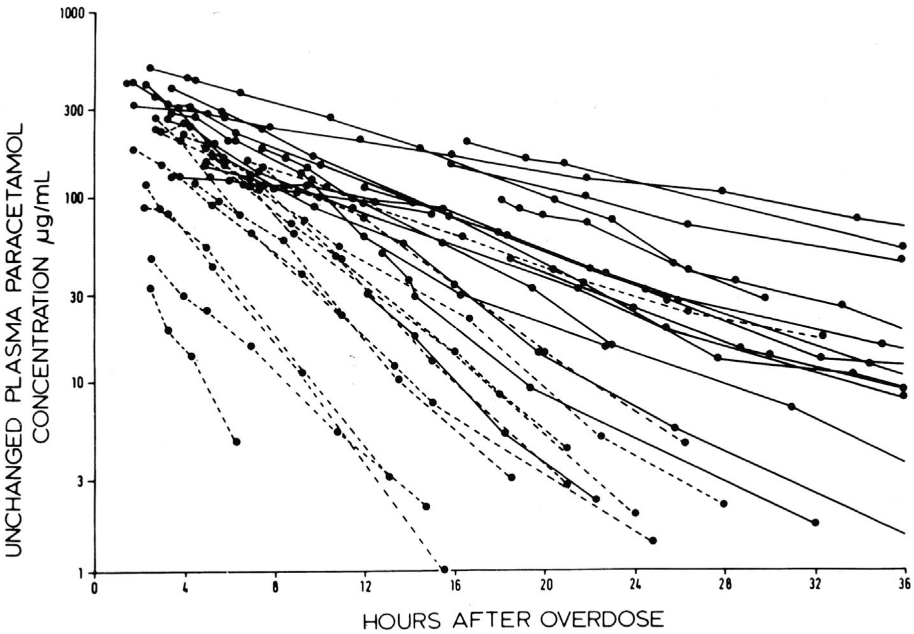

FIGURE 1 Plasma concentrations of paracetamol in 30 patients without liver damage (dashed lines) and with liver damage (continuous lines) following overdosage. The patients were admitted before effective treatment became available [7].

3 | THE RUMACK–MATTHEW NOMOGRAM

In 1973 Barry Rumack was attached to our unit as a visiting scientist and, with Henry Matthew, he later drew up a nomogram for prediction of liver damage, which was based on Edinburgh data.[18] In the Rumack–Matthew nomogram, plasma paracetamol concentrations taken at 4 h or later after the overdose were plotted on a logarithmic scale against time and related to a line starting at 200 mg/L at 4 h after ingestion with the concentration halving every 4 h. Depending on the concentrations chosen for the 4-h time point, patients could be stratified into discrete zones of risk but the slope of the line always corresponded to a 50% decrease in concentrations every 4 h. Apparently, the choice of 4 h had nothing at all to do with the kinetics of paracetamol in poisoned patients and was drawn instead to discriminate between patients with and without liver damage.[19] This was an inspired choice and it came down to the same thing anyway as the paracetamol half-life of 4 h represents the critical threshold for hepatotoxicity in poisoned patients.[15] In patients destined to suffer severe liver damage, plasma paracetamol concentrations will rise progressively above the line with time while those in which it was <4 h will fall progressively below the line with time. A crucial element to the Rumack–Matthew nomogram is the slope representing a halving of the paracetamol concentrations every 4 h.

The introduction of the Rumack–Matthew nomogram was a major step forward and it is used widely as a guide to the severity of poisoning and the need for treatment. Adult patients with a plasma paracetamol concentration above a line starting at 200 mg at 4 h after ingestion will have absorbed >150 mg/kg of paracetamol and before effective treatment became available, about 60% developed severe liver damage. Above a parallel line starting at 300 mg at 4 h there was a 90% chance of severe liver damage while the risk was very low below the line starting at 100 mg.[13] The positioning of the nomogram line as the guide for treatment is inevitably a compromise. If set too high, many patients will not be treated and will suffer severe and possibly fatal liver damage. But if the line is too low, many patients will be treated unnecessarily with large increases in admissions, costs and in the proportion suffering adverse reactions to *N*-acetylcysteine (NAC).[20] There are other limitations with the nomogram. It depends crucially on the patient giving the correct time of the overdosage and *line crossing* from below to above the line can occur if absorption is delayed when a slow-release formulation has been taken or following the combined ingestion of drugs that slow gastric emptying such as opiates or anticholinergics.[21] In addition, there may be difficulties with staggered (repeated) overdoses. According to a recent survey, the nomogram is not always used correctly and the use of an online calculator has been recommended.[22] In the UK, the nomogram treatment line was initially set at 200 mg/L at 4 h[23] while, under pressure from the Food and Drug Administration, a lower line corresponding to 150 mg/L at 4 h was used in the USA.[18,24] In 2012, on the recommendation of the UK Commission on Human Medicines the treatment line was lowered to 100 mg/L at 4 h. As a result, there was a large increase in admissions, many more patients were treated with NAC and more suffered adverse reactions to it.[25]

The predictive value of the Rumack–Matthew nomogram could be greatly increased by also taking into account the paracetamol half-life. The concentration–time data could be entered into a simple online program or smartphone App, which would calculate the half-life, factor in the position of the data points relative to the nomogram treatment line, apply corrections for risk factors, and provide the optimum intensity and duration of treatment with NAC. This combination would be a very useful predictor of the outcome and guide to treatment. For example, if the paracetamol concentrations are below a treatment line starting at 150 mg/L at 4 h, and the half-life is <4 h, the risk of significant hepatotoxicity is negligible and the patient could be discharged medically with no need for follow up. Conversely, if the paracetamol concentrations are above the 300 mg/L line on the nomogram and the half-life is 10 h or more there will be very severe liver damage.

4 | DISCOVERY OF THE MECHANISMS OF HEPATOTOXICITY

The mechanism of paracetamol hepatotoxicity was discovered fortuitously. In 1971, I attended a lecture in the Department of Pharmacology, University of Dundee given by Bernard Brodie, a pioneering biochemical pharmacologist from the National Institutes of Health, Bethesda. He described a recent visit to Australia where he stayed with an old friend who was a sheep farmer. His friend had told him about the difficulties he was having with treating his sheep for worms with carbon tetrachloride. While most seemed to be unaffected, others died in a random pattern with severe liver damage. Carbon tetrachloride is a well-known hepatotoxin but, at the same time, it is inert and chemically inactive, so much so that it was formerly used extensively as a fire extinguisher. Brodie was intrigued by this and on his return to Bethesda, together with postdoctoral student Jerry Mitchell, he carried out studies on bromobenzene, another hepatotoxic halogenated hydrocarbon which produced typical centrilobular hepatic necrosis. They showed that, in mice, bromobenzene was bound covalently to liver proteins and excreted in the urine as glutathione conjugates. Furthermore, bromobenzene was converted by hepatic cytochrome P450 enzymes to a reactive epoxide metabolite, which, after a small nonhepatotoxic dose, was removed safely by conjugation with hepatic glutathione, but, after a large toxic dose, glutathione was depleted and the epoxide metabolite was bound covalently to liver enzymes and proteins causing hepatic necrosis. Brodie described how hepatotoxicity was increased or decreased by treatments that increased or decreased the activity of cytochrome P450 drug-metabolizing enzymes. Similarly, the severity of liver damage was increased by the administration of agents that depleted hepatic glutathione and decreased by treatments that stimulated its synthesis.[26] All the time that I was listening to Brodie giving his lecture, my thoughts kept returning to the problems of paracetamol overdosage. Could a similar mechanism explain the hepatotoxicity of paracetamol? After the lecture I had a long discussion with Bernard in the bar of the Dundee hotel where he was staying. He was unaware of the problems of liver damage following an overdose of paracetamol and was immediately very interested. He promised to investigate urgently the mechanism of paracetamol hepatotoxicity on his return to the National Institutes of Health. Subsequently, in an elegant series of studies, Jerry Mitchell and his colleagues showed that indeed, the mechanism of paracetamol hepatotoxicity was very similar to that of bromobenzene. It was converted by P450 enzymes to a highly reactive minor metabolite, later shown to be N-acetyl-p-benzoquinoneimine, which was normally inactivated by conjugation with hepatic glutathione. However, as with bromobenzene, large doses of paracetamol caused depletion of hepatic glutathione, covalent binding and acute hepatic necrosis. The factors that modified the toxic response to bromobenzene were shown to have the same effects on the hepatotoxicity of paracetamol. Once again, protection against hepatotoxicity depended crucially on the availability of hepatic glutathione. The rate of synthesis of glutathione is limited by the availability of inorganic sulfate for the synthesis of l-cysteine and this could be maintained by the administration of sulfydryl compounds such as l-cysteine, methionine and cysteamine. Crucially, these agents were very effective in protecting against paracetamol hepatotoxicity in mice, even when administration was delayed.[27–30] I was in frequent contact with Jerry Mitchell over this period and at last the way was open for the introduction of effective treatments for paracetamol poisoning.

5 | THE FIRST EFFECTIVE TREATMENT: CYSTEAMINE

Any treatment would have to be given as early as possible because at that time we believed that most of the toxic metabolite would be formed during the first few hours when the paracetamol concentrations were highest. It would also have to be administered intravenously as, in our experience, severely poisoned patients invariably developed protracted vomiting within a few hours of the overdose. Cysteine and methionine were the obvious candidates but they had limited aqueous solubility and required the rapid infusion of very large volumes of fluid. We decided therefore to use cysteamine, which was very water soluble and had had some very limited clinical use for radioprotection. It was not available as a pharmaceutical product and we had to use standard laboratory reagent, which was sterilized immediately before use by passing through a bacterial filter. But first we had to determine the maximum tolerable dose. In those days, there were no Ethics Committees or Institutional Review Boards to regulate clinical research, so my registrar and I gave ourselves increasing doses of cysteamine. We soon made ourselves very ill and I had to be admitted to my own medical ward as an emergency. Cysteamine was awful. It smelt like rotten cabbage and as the dose increased it produced intense flushing, hot sweats, tachycardia, continuous vomiting and eventually incapacitation. We devised an arbitrary dose regime based on the rapid administration of what we considered to be the maximum tolerable dose followed by a decreasing rate of administration over a period of 20 h. The first use of cysteamine was restricted to patients who were admitted with such high plasma paracetamol concentrations (above the 300 mg/L at 4 h line on the nomogram) that severe liver damage was inevitable. In addition, because we thought that cysteamine could only be hepatoprotective during the early phase of the metabolic activation of paracetamol, its use was restricted to patients who could be treated within 8 h of the overdose.

The first patient to receive cysteamine said she had taken 100 tablets of paracetamol 3 h and 45 min before admission. Severe liver damage would have been inevitable as the plasma paracetamol concentration was 351 mg/L and treatment with cysteamine was started 5 h and 30 min after the overdose was taken. Amazingly, there was no liver damage. The same patient was admitted on 2 further occasions with paracetamol concentrations of 470 and 365 mg/L at 3 h 45 min and 6 h 30 min. Cysteamine was commenced 4 h 15 min and 8 h 15 min after the overdoses, respectively. Liver function tests remained normal on the first occasion, and on the second there was only a minimal transient elevation of the transaminases. Four other patients with very high paracetamol levels were treated with cysteamine within 10 h. Serial liver function tests remained entirely normal in 3 and in the fourth patient there was only a minor transient increase of the plasma alanine aminotransferase activity to 192 IU/L.[23] Subsequent studies confirmed the remarkable efficacy of cysteamine in preventing liver damage in severely poisoned patients treated early but protection was lost when treatment was delayed >10 h after ingestion.[31]

The unpleasant adverse effects of cysteamine together with its lack of availability as a pharmaceutical product prompted the search for alternatives. Intravenous cysteine and methionine were as effective as cysteamine in preventing liver damage after paracetamol overdose and they did not cause significant adverse effects.[31] However, there were practical problems with IV administration due to their limited solubility. Oral methionine was used successfully but there were some failures.[32]

6 | *N*-ACETYLCYSTEINE

We first suggested the use of NAC in 1974 and it is essentially a water-soluble form of cysteine.[33] Furthermore, it was already on the market in the UK as a sterile solution for intrabronchial administration (Airbron). Fifteen poisoned patients with admission plasma paracetamol concentrations above a treatment line starting at 200 mg/L at 4 h on the nomogram were treated with IV NAC. The mean 4-h plasma paracetamol concentration was 369 mg/L. Liver function tests remained normal or were only slightly disturbed in 11 of 12 patients treated within 10 h but severe liver damage occurred in the other patient and in 3 in whom treatment was delayed for >10 h. NAC was given IV as an initial dose of 150 mg/kg over 15 min followed by 50 mg/kg in 4 h and 100 mg/kg over the next 16 h making a total of 300 mg/kg in 20 h 15 min. It was well tolerated and no serious adverse effects were noted.[34] There has since been speculation about the rationale for this regime.[35] It was simply that at the time, we thought that there would only be a small window of opportunity as any antidote could only be effective if given very rapidly and early to limit glutathione depletion before the main part of the paracetamol had been metabolized. It is now obvious that for a limited time only, these sulfydryl compounds can reverse postarylation covalent binding.

Further studies confirmed the remarkable efficacy of IV NAC for the treatment of paracetamol overdose. It also protected against renal failure and death but again there was a critical ingestion to treatment interval of 8 h, after which time efficacy diminished progressively. There was no evidence of protection after 15 h,[36] although NAC was of benefit by a different mechanism if given later to patients with established hepatic failure.[37] The importance of early treatment was shown in the case of a patient who was admitted to our unit for self-poisoning 82 times over 6 years. Paracetamol was taken on 31 occasions and poisoning was usually severe with paracetamol concentrations far above the standard treatment nomogram at 200 mg/L at 4 h, and urinary recovery of up to 20 g of the drug despite late admissions and incomplete collections. The patient was treated with IV cysteamine, methionine, cysteine and NAC on 13 admissions and never suffered severe liver damage when treatment was started within 12 h of taking the overdoses. But with later presentations, acute liver injury occurred at least 7 times.[38]

Meantime, the efficacy of oral NAC (Mucomyst) was demonstrated in several large American multicentre studies.[39–41] The dose was 140 mg/kg followed by 70 mg/kg every 4 h for 17 doses giving a total of 1330 mg/kg. In a report on 11 195 patients, 2023 had paracetamol concentrations above a lower treatment line joining 150 mg/L at 4 h and 5 mg/L at 24 h. Early treatment with oral NAC was very effective in preventing liver damage and death but these occurred increasingly as treatment was delayed beyond 10 and 16 h.[41] There has been much discussion concerning the superiority or otherwise of oral *vs*. IV NAC but both routes are undoubtedly effective if given early.[42–44] The oral route has the advantage that the whole absorbed dose is delivered directly to the liver, but administration can be impractical in severely poisoned patients who are vomiting.

7 | ADVERSE REACTIONS TO NAC

NAC can occasionally cause anaphylactoid reactions characterized by nausea, vomiting, flushing, urticaria and pruritis. These dose-dependent effects occurred most often during the first hour of treatment with the original 20-h IV regime when concentrations of NAC were highest. They usually subsided rapidly on slowing or stopping the infusion and were probably related to release of histamine. More severe reactions include bronchospasm, angioedema and hypotension.[45,46] To reduce the risk of these early anaphylactoid reactions, we modelled alternative regimes to give lower NAC concentrations with slower initial rates of infusion, but the same total dose of NAC given over 20 h.[47] But in the late 1980s there was an Ethics Committee at the Edinburgh Royal Infirmary and when we submitted a protocol for the reduction of the initial rate of infusion while maintaining the same total dose it was refused on the grounds that the existing regime was highly effective and that any change might be less effective!

8 | CHANGES IN DOSE AND DURATION OF TREATMENT

It is now 50 years since the first effective treatment of paracetamol overdose and during this time there have been important developments in management. However, opinion is divided concerning the optimum dose and duration of treatment with NAC, especially for patients who present late.[35] Simpler and shorter regimes for the IV administration of NAC with slower initial rates of infusion have at last been introduced resulting in a reduced incidence of adverse effects.[20,25,35,48–52] A new high-dose single bag IV NAC regimen was associated with lower peak transaminase and fewer patients becoming hepatotoxic as compared to the standard 3-bag 20-h regimen.[53] In another report, NAC was discontinued safely after 12 h in selected low risk patients.[54] By contrast, patients who have taken *massive* overdoses of paracetamol are at increased risk of liver damage even when treated early with NAC.[55] Conditions that could possibly increase the risk of hepatotoxicity apart from late presentation include staggered overdoses, age, ethnicity, obesity, weight loss surgery, fasting, starvation, poor nutrition, the use of enzyme inducing drugs, chronic alcoholism and debilitating diseases.[19,56–63] It has rightly been suggested that these and severely poisoned patients require larger doses of NAC and a longer duration of treatment.[64–66] In 1 report of patients taking massive paracetamol overdoses, treatment with increased doses of NAC was associated with a significant reduction in hepatotoxicity.[67]

9 | RECENT ADVANCES

The aminotransferases have been the traditional markers of paracetamol hepatotoxicity but currently there is much interest in novel biomarkers such as microRNAs, high mobility group box-1, keratin-18, procalcitonin and cyclophilin which promise enhanced liver specificity and sensitivity to provide an earlier prognosis.[68–70] In addition, cytochrome P450 metabolites and paracetamol-protein adducts are specific biomarkers of toxic paracetamol metabolite exposure[71,72] and a point-of-care immunoassay has been developed for their rapid detection.[73] Another approach to early assessment of the severity of poisoning would be the monitoring of plasma concentrations of cysteine and/or inorganic sulfate. Liver damage does not occur until there is 70% depletion of hepatic glutathione[30] and there is competition for inorganic sulfate for the synthesis of glutathione and the sulfate conjugation of paracetamol. Reduced sulfate conjugation leads to a shift in biotransformation to toxic oxidation pathways and patients with low sulfate reserves are at a higher risk of paracetamol toxicity.[74]

Over the years, there have been important advances in the management of paracetamol poisoning and early treatment with NAC is very successful in preventing severe liver damage and death. But apart from the last resort of liver transplantation in patients who develop liver failure, there is still no effective treatment for patients who present late. There have been further developments in treatment based on different mechanistic approaches, which give hope for effective late treatment when NAC is no longer effective.[75,76] Studies are in progress with calmangafodipir, an agent that mimics the actions of superoxide dismutase[77] and 4-methylpyrazole (fomepizole), which not only inhibits the metabolic activation of paracetamol but also specifically inhibits c-Jun N-terminal kinase activation which is critically involved in paracetamol hepatotoxicity.[78]

Many years ago, it was suggested that methionine should be added to paracetamol tablets as this prevented hepatotoxicity in animals.[79] This idea was abandoned because vast numbers of responsible users of paracetamol would have no choice but to take it with methionine to protect the tiny minority taking an overdose.[80] Paracetamol poisoning imposes a very heavy burden on health services worldwide, not to mention the costs, and deaths still occur. Cysteine reacts with reducing sugars in a Maillard reaction to yield meaty and brown food flavours and it is freely available not only as a food additive (E920) but also as a nutritional supplement (in tablets of 500 and 600 mg). Perhaps the time has come to solve the problems of paracetamol overdose at a stroke by adding a small amount of cysteine to all paracetamol tablets.

CONFLICT OF INTEREST STATEMENT

None.

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